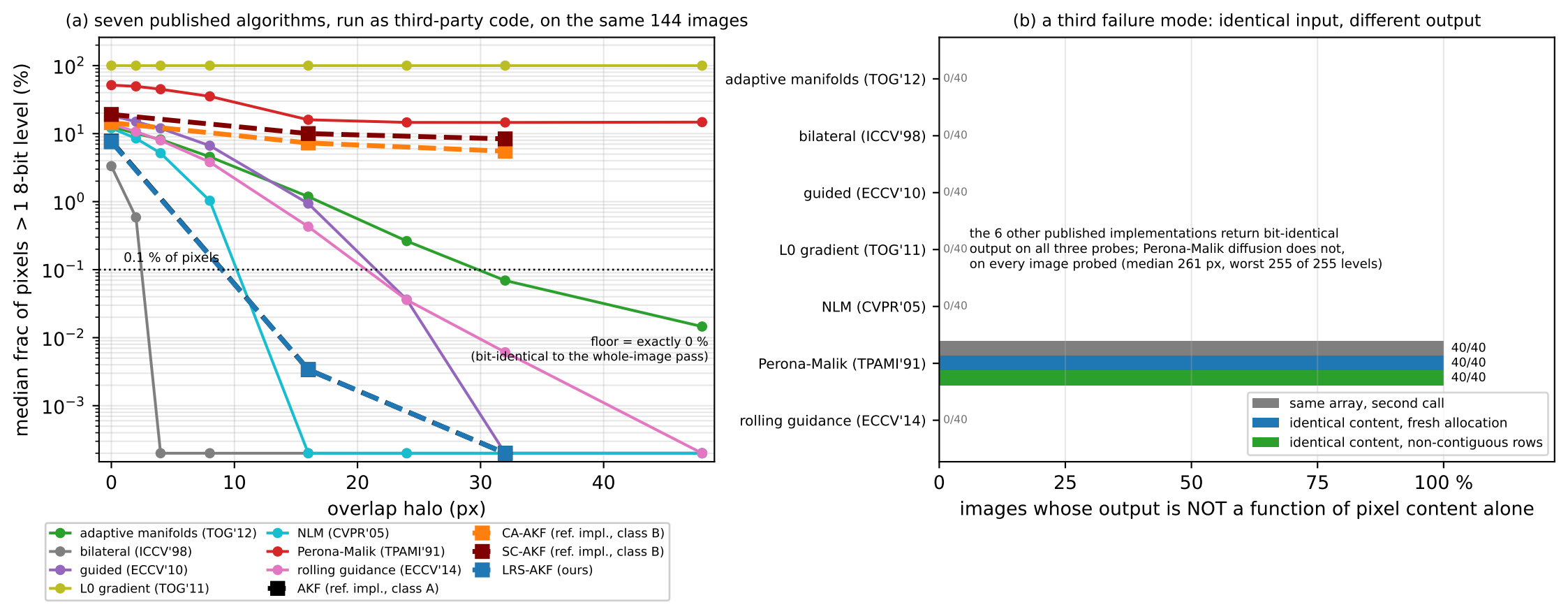
(a) seven published algorithms, run as third-party code, on the same 144 images
median frac of pixels > 1 8-bit level (%)
overlap halo (px)
0.1 % of pixels
floor = exactly 0 %
(bit-identical to the whole-image pass)
adaptive manifolds (TOG'12)
bilateral (ICCV'98)
guided (ECCV'10)
L0 gradient (TOG'11)
NLM (CVPR'05)
Perona-Malik (TPAMI'91)
rolling guidance (ECCV'14)
AKF (ref. impl., class A)
CA-AKF (ref. impl., class B)
SC-AKF (ref. impl., class B)
LRS-AKF (ours)
(b) a third failure mode: identical input, different output
0/40
the 6 other published implementations return bit-identical
output on all three probes; Perona-Malik diffusion does not,
on every image probed (median 261 px, worst 255 of 255 levels)
40/40
same array, second call
identical content, fresh allocation
identical content, non-contiguous rows
images whose output is NOT a function of pixel content alone